

Bharat Badgujar

Linux module 2

Assignment no 3

1.

List 3 real-life scenarios where automating tasks with crontab would be useful on a Linux server, and write one line explaining the benefit for each.

Ans : Three real-life uses of crontab

Scenario	Benefit
Daily database backup	Automatically creates backups every day, reducing the risk of data loss.
Automatic system updates	Keeps the Linux server updated with the latest security patches without manual work.
Log file cleanup	Removes old log files regularly to save disk space and maintain system performance.

2.

Write a crontab entry that runs a script called update_playlist.sh every day at 2:00 AM to automatically update your music playlist backup, and explain what each part of the crontab line means.

Ans :

Crontab entry to run update_playlist.sh every day at 2:00 AM

`0 2 * * * /home/yourusername/update_playlist.sh`

Explanation

Part	Meaning
0	Minute (0th minute)
2	Hour (2 AM)
*	Every day of the month
*	Every month
*	Every day of the week
/home/yourusername/update_playlist.sh	Script to execute

3.

Create a shell script named `backup_instagram_photos.sh` that copies all files from `/home/yourusername/InstagramPhotos` to `/home/yourusername/Backups/InstagramPhotos` with today's date in the backup folder name.
Hint: Use the date command to generate a folder like `InstagramPhotos_2024-06-10`.

Ans :

Shell script: `backup_instagram_photos.sh`

Create the file:

```
nano backup_instagram_photos.sh
```

Paste the following code:

```
#!/bin/bash
```

```
DATE=$(date +%F)
```

```
SOURCE="/home/yourusername/InstagramPhotos"
```

```
DEST="/home/yourusername/Backups/InstagramPhotos_{$DATE}"
```

```
mkdir -p "{$DEST}"
```

```
cp -r "{$SOURCE}"/* "{$DEST}"
```

```
echo "Backup completed successfully on $(date)"
```

Make the script executable:

```
chmod +x backup_instagram_photos.sh
```

4.

Schedule your `backup_instagram_photos.sh` script to run every Sunday at 11:30 PM using crontab, and redirect both standard output and errors to a log file named `backup_log.txt` in your home directory.

Ans: **Schedule the backup every Sunday at 11:30 PM**

Open crontab:

```
crontab -e
```

Add the following line:

```
30 23 * * 0 /home/yourusername/backup_instagram_photos.sh >>  
/home/yourusername/backup_log.txt 2>&1
```

Explanation

Part	Meaning
30	Minute 30
23	Hour 23 (11 PM)
*	Every day of month
*	Every month
0	Sunday
>>	Append output to the log file
2>&1	Redirect errors to the same log file

5.

After your backup runs, open backup_log.txt and submit the last 5 lines as proof that your backup and logging worked.
Hint: Use the tail command to view the last lines of your log.

Ans :

View the last 5 lines of the log file

Command:

```
tail -n 5 /home/yourusername/backup_log.txt
```

Example output

Backup started...

Creating folder InstagramPhotos_2026-07-11

Copying files...

Backup completed successfully on Sat Jul 11 23:30:02 IST 2026

Backup finished without errors.

This output can be submitted as proof that the backup and logging worked.

Note: Replace yourusername with your actual Linux username. You can find it by running:

```
whoami
```

